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1 PURPOSE

This Standard Operating Procedure (SOP) describes the techniques used to measure the titres of total human immunoglobulin G (IgG) responses against a full length recombinant clamped MERS-CoV spike glycoprotein (GP) in human plasma and serum samples. The measured antibody titres will serve to indicate the humoral immune response to a candidate ChAdOx1 MERS vaccine encoding the full length spike GP of the MERS-CoV in human subjects.

This assay must be performed in accordance with the principles of the International Conference on Harmonisation Tripartite Guideline on Good Clinical Practice and the Human Tissue Act. Informed consent must be in place for samples to be tested in this assay.

2 INTRODUCTION

One of the primary readouts for assessing the protective humoral immunity against this candidate vaccine is the measurement of total IgG responses to MERS-CoV spike GP in the blood of vaccinated volunteers. This will be done as described in this SOP using an in-house ELISA that includes a standard curve derived from a pool of the volunteers' sera containing highly concentrated anti-MERS IgG.

3 SCOPE

This SOP applies to all studies sponsored by the University of Oxford that are conducted by the Jenner Institute where measurement of total IgG against the candidate ChAdOx1 MERS vaccine is required, for which ELISA is the primary readout.

4 DEFINITIONS/ABBREVIATIONS

ELISA = Enzyme-Linked ImmunoSorbent Assay.

MERS-CoV = Middle East Respiratory Syndrome Coronavirus.

ChAdOx1 MERS = replication-deficient simian adenovirus vector ChAdOx1, expressing a codon-optimised coding sequence for the full-length spike GP of the MERS-CoV isolate Camel/Qatar_2_2014 (GenBank, accession number KJ650098.1), including a 32 amino acid N-terminal tissue plasminogen activator leader sequence.

5 RESPONSIBILITIES

All staff employed by University of Oxford who work on outbreak pathogen vaccine trials including clinicians or visiting scientists working in the lab, must follow this protocol.

Senior immunologist is responsible for training new members of staff to be competent with this SOP and ensuring that sample receipt is conducted according to this SOP

6 EQUIPMENT/REAGENTS

6.1 Equipment:

Fridge (+4°C)



Freezer (-20°C and -80°C)

Vortexer

Eppendorf Racks

Pipettes including 8- or 12-well multi-channel and automatic multi-channel

Pipetteboy and stripettes

Bio-tek ELx800 Microplate Reader with Gen5 ELISA software

Timer

Sufficient tips for pipettes. 0.1-10 µL, 2-20 µL, 20-200 µL, 100-1000 µL

Safety Glasses

PBS-Tween Immunowash

6.2 Reagents

Reagent	Company	Cat #
NUNC Immuno Plates (442404)	Fisher	DIS-971-030J
Blocker Casein in PBS	Thermo Scientific	37528
Goat anti-human IgG (γ-chain)-alk phos	Sigma	A3187
Dulbecco's PBS (DPBS)	Sigma	D8537
5x Diethanolamine Buffer	Pierce	34064
4-Nitrophenyl Phosphate Tablets (20 mg)	Sigma	N2765
Tween-20	Sigma	P7949
1.5 mL Eppendorf tubes	Fisher	FB74031
Reagent Reservoirs (Costar 4870)	Fisher	PMP-331-010C
Water (sterile-filtered, Bioreagent)	Sigma	W3500
DPBS powder (10 L)	Invitrogen	21600-069
MERS-CoV clamped spike GP (0.5 mg/mL)	-	-

6.3 Buffers and solutions:

DPBS Use for coating plates with MERS CoV Spike Glycoprotein.

DPBS/T (10L) (DPBS with 0.05% Tween) for washing plates. Dissolve one DPBS powder tub in 10 L deionised water (15.0 MΩ setting). Add 5 mL Tween-20. Shake and return to the ELISA plate wash station.

Blocking and Dilution buffer Blocker Casein in PBS

Anti-human IgG Alkaline phosphatase 1:1000 dilution in casein. 6 µL in a final volume of 6 mL (6 mL per plate).

Developer Dilute 5X Diethanolamine (DEA) substrate buffer to 1X in sterile-filtered, Bioreagent Water. Add one 20 mg pNPP tablet per 20 mL. 10 mL is required per plate (minimum that can be made up is 20 mL). Make up just before use, wrap in foil and vortex well.



7 PROCEDURE

7.1 Day 1

7.1.1 Coating ELISA plates on the bench

- Print off a new ELISA record sheet for each experiment (accessed from: X:\KEwer\7. ELISA\Templates and protocols).
- Number the experiment with the next experiment number and fill this in with the required information throughout the experiment.
- Calculate the number of Nunc Immuno ELISA plates required (One plate: 23 samples and an internal control in triplicate in addition to the standard curve in duplicate)
- Thaw an aliquot of MERS-CoV clamped spike GP (typically 0.5 mg/mL stock solution kept at -80°C and prepare as shown in the table below. Calculate accordingly if stock protein concentration differed). Coat plates with 50 µL/well of MERS CoV Spike clamped GP at 1.0 µg/mL in dPBS). Cover plates in clingfilm and store over-night for 18 h at 4°C. Note time of coating and record on the experiment record sheet.

Number of plates to coat	Volume of Spike GP (0.5 mg/ml stock solution)	Volume of DPBS
1	10 µL	4.99 mL
2	20 µL	9.98 mL
3	30 µL	14.97 mL
4	40 µL	19.98 mL
5	50 µL	24.95 mL
6	60 µL	29.94 mL

7.2 Day 2

7.2.1 Blocking plates

- Thaw required serum samples (up to 23 samples per plate) at room temperature (RT) for at least an 1 h.
- Flick off coating solution into the sink (wear eye protection).
- Wash plates 6x in DPBS-T using the handheld plate washer and tap dry.



- Block plates with 100 μL per well of casein.
- Stack plates (with an empty blank plate on top). Leave for 1 h at RT. Note time of blocking and record on the experiment sheet.
- During blocking prepare serum samples and reference standard dilutions as below (7.2.2).
- After 1 hour incubation for blocking, flick off blocking solution into the sink and tap dry.

7.1.2 Sample/standard preparation

- Vortex the thawed samples thoroughly and dilute in casein at 1:500 (5 μL sample diluted in 2495 μL casein and vortexed). All sample dilutions MUST be done using a minimum of 5 μL sample.
- Add 50 μL of diluted sample to each well (each sample in triplicate). Store diluted serum overnight in fridge in case repeat of samples at higher dilution is necessary.
- The standard curve is a two-fold dilution series with an initial dilution of 1:100 from the positive standard pool. To avoid multiple freeze-thaw of the positive standard pool, prepare aliquots of 1:100 working standards in casein to a final volume of 600 μL (6 μL pool sample in 594 μL casein) and store in -80°C freezer, where it is stable for up to a year.
- All standard dilution is done using Eppendorf tubes labelled Tube1 – Tube10. To run 2 plates prepare standards as follows: Add 520 μL of the 1:100 working standard to the first tube (Tube1). Add 260 μL of casein to the rest of the tubes (Tube2-Tube10). Transfer 260 μL from Tube1 to Tube2 (mix 15 times). Transfer 260 μL from Tube2 into Tube3 and continue across the rest of the tubes. Discard 260 μL from the last tube (Tube10).
- Make a 1:800 dilution of the standard serum pool in casein to use for the plate internal control. (eg 175 μL casein + 25 μL positive standard pool per plate). Add to wells D12, E12 and F12.
- If one plate is assayed, refreeze the remaining 1:100 working standard serum and reuse in subsequent assays. This can be freeze-thawed up to three times if required.

7.1.3 Plating out

- After the blocking incubation period, flick off the blocker casein and tap dry the plates.
- Add 50 μL of the thoroughly vortexed diluted samples (in triplicate) and the blank control (casein only) as shown in the plate layout below using an automatic pipette if available. Without adding the standards and internal control, cover plate with an empty blank plate and repeat step if multiple plates are prepared.



Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10	Sample 11	Sample 12
Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10	Sample 11	Sample 12
Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10	Sample 11	Sample 12
Sample 13	Sample 14	Sample 15	Sample 16	Sample 17	Sample 18	Sample 19	Sample 20	Sample 21	Sample 22	Sample 23	Internal control
Sample 13	Sample 14	Sample 15	Sample 16	Sample 17	Sample 18	Sample 19	Sample 20	Sample 21	Sample 22	Sample 23	Internal control
Sample 13	Sample 14	Sample 15	Sample 16	Sample 17	Sample 18	Sample 19	Sample 20	Sample 21	Sample 22	Sample 23	Internal control
St 1:100	St 1:200	St 1:400	St 1:800	St 1:1600	St 1:3200	St 1:6400	St 1:12800	St 1:25600	St 1:51200	Blank	Blank
St 1:100	St 1:200	St 1:400	St 1:800	St 1:1600	St 1:3200	St 1:6400	St 1:12800	St 1:25600	St 1:51200	Blank	Blank

- Add 50 μ L of the 1:100 Standard as shown in the plate layout to all plates first then followed by 1:200 and so on. Add the internal control immediately after adding the 1:800 standard then followed by the rest of the standards. Use an automatic pipette if available for transferring the standards.
- Incubate plate at RT for 2 h with a blank plate on top.

7.1.4 Secondary Antibody

- After 2 h incubation with samples, wash plates 6x in DPBS-T using the handheld plate washer and tap dry.
- Dilute anti-human IgG antibody alkaline phosphatase 1:1000 in casein (see section 6.3).
- Add 50 μ L of detection antibody to each well.
- Incubate plate at RT for 1 h with a blank plate on top.

7.1.5 Development and plate reading

- Prepare the pNPP substrate (development buffer) as shown in the table below. Make up the substrate during the antibody incubation, at least 30 min prior to development to allow adequate time for the pNPP tablet time to dissolve and protect from light until use (see section 6.3).



Number of plates to develop	Number of pNPP tablets required	Volume of 5X DEA buffer required	Volume of H ₂ O
1	1	4 mL	16 mL
2	1	4 mL	16 mL
3	2	8 mL	32 mL
4	2	8 mL	32 mL
5	3	12 mL	48 mL
6	3	12 mL	48 mL

- After 1 h incubation with secondary antibody, wash plates 6x in DPBS-T using the handheld plate washer and tap dry.
- Add 100 µL of development buffer to each well.
- Turn the plate reader on and log in to the computer.
- Open the Gen5 software, create new experiment using the protocol template (MERS STANDARDISED ELISA_SAMPLE ANALYSIS TEMPLATE) located in this folder (X:\Vaccine\10. MERS-CoV\1. MERS001_Oxford\12.0 Immunology\8. ELISA\Data and analysis\MERS001 STANDARDISED ELISA_SAMPLE ANALYSIS DATA).
- Start taking readings of the plate at OD405 after 10 minutes and repeat readings until the control readings and curve parameters fit expected values.
 - internal control OD (wells D12, E12, F12) should be around 0.9-1.1 (1300-2300 EU).
 - St 1:100 wells (G1 and H1) should be 3.5-3.8
 - St 1:800 wells (G4 and H4) should be close to the values in the internal control wells (D12, E12, F12).
 - Blank wells should be <0.2
 - Standard curve reaches 4 parameter ranges below:

Parameters	A	B	C	D	R ²
Ideal range	0.1-0.2	1-1.3	5-13	4.5-6	>0.996



- Export the plate data, standard curve and curve results to Excel. Record time of reads. Save the Gen5 Experiment and the Excel worksheet in the corresponding experiment folder. Trial specific assay data must be stored in the corresponding clinical trial folder on the X drive or for blinded studies use X:\KEwer\7.ELISA\ 14. Confidential Immunology Data.

7.1.6 Data Analysis

- Analyse the data in the Excel sheet “ELISA analysis template” (accessed from: X:\KEwer\7.ELISA\Templates and protocols). All blue-shaded fields in the worksheets MUST be completed or confirmed as correct.
- Fill in the assay information and plate layout in the blue boxes in the “Plate layouts and raw data” page, then paste the plate layout, plate data and parameters (A,B,C,D and R²) into the “plate layout and data” sheet.
- Check for outliers (“high CV” on Results page) and take average of duplicates instead of triplicates if necessary. On the Raw data, highlight outlying well in yellow, calculate the average of the other two wells and type this into the yellow outlier cell.
- The ELISA units for each sample are on the Results sheet.
- Individual plate data and QC can be seen on the individual plate sheets.
- The sheet will indicate if a sample dilution is too high or low. If the sheet indicates that sample dilution is “TOO LOW” at 1:100, the sample is negative and can be assigned an arbitrary value of 1.
- If the sheet indicates that sample dilution is too low at 1:500 repeat at 1:100. For samples that are too high at 1:500 repeat them at higher dilution e.g. 1:1000, 1:2000 or more. When analysing samples that have been retested at 1:100 or higher dilutions such as 1:1000, change the dilution in the blue boxes on the individual plate pages (“plate 1”, “plate 2”, “plate 3” etc.)
- If further dilutions from the 1:500 dilutions are not required then the tubes can be discarded.
- Samples with values greater than 225 ELISA units are positive.
- All assay parameter data for each experiment must be added to a designated QC workbook.
- All ELISA data for clinical trials must be QC’d prior to publication.

8 RELATED DOCUMENTS

J284 Standardised ELISA set-up
ELISA analysis template.xls
ELISA Record sheet.pdf



9 REFERENCES

ICH Harmonised Tripartite Guideline for Good Clinical Practice
MHRA Guidelines 'Good Clinical Practice for Clinical Laboratories

10 TRAINING

Record the level of training required by users

Training type	Required (tick at least one box)
Read and Understand	YES
Attend face to face training based on SOP	YES
Competency assessment	YES
List other specific training types if required	

Read and understand training means that the user will be sufficiently trained to perform the activity described after reading the SOP and having any questions answered by the author or other trained individual.

Attend face to face training based on SOP training means that the user must read and understand the SOP, but must also attend face to face training. This enables additional discussion to be undertaken and clarification where necessary. The trainer is able to emphasise specific requirements that may differ from standard practice. This may often be performed as a group session.

Competency assessment requires the user to read and understand the SOP and also to receive face to face training (usually 1:1 or small groups). Specific practical aspects of the procedure will be covered and the individual must be able to demonstrate competency in all aspects prior to being signed off by the trainer.

11 SAFETY

This section should list any specific health and safety issues with including reference to COSHH, risk and VDU assessments that are identified as **medium** or **high** risk.

The risk assessment template below (appendix 1) should be completed for all procedures that are assessed as requiring a full risk assessment. This will usually be all clinical procedures and some office based activities. A trained risk assessor should review each new SOP to determine if a full risk assessment is required.

Some risk assessment templates are available on request, however, these must always be adapted for the specific procedure.

The risk assessment should be conducted by the author or reviewer of the SOP and a trained risk assessor.

Risk assessment required No

12 REVIEW HISTORY

Record the changes made from the previous version in the table below.

Date	Reviewed By	Revision type (New / Major / Minor)	Detail (significant changes from previous version)	Previous Version



JENNER VACCINE TRIALS

NUFFIELD DEPARTMENT OF MEDICINE



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